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POWER SYSTEM FOR MARITIME VEHICLE

Abstract

A retention assembly for retaining a power source in a maritime vehicle includes at least one strap member and a buckle. The at least one strap member includes a first end, a second end, and an elongated length therebetween. The buckle is operably coupled with a portion of the elongated length of the at least one strap member and includes a quick release mechanism movable between a first, disengaged position and a second, engaged position. The second, engaged position is adapted to generate a tension in the at least one strap member to retain the power source within a cavity.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/561,063, filed on Mar. 4, 2024, the entirety of which is herein expressly incorporated by reference.

FIELD OF THE DISCLOSURE

[0002] This disclosure relates generally to maritime vehicles, and, more particularly, to approaches for retaining and releasing a power source in a maritime vehicle.

BACKGROUND

[0003] Maritime vehicles, or vehicles designed for use on or in the water, are commonly used for transportation, recreation, defense, scientific research, and other purposes. Examples of maritime vehicles include boats, watercraft, submarines, and amphibious vehicles. Maritime vehicles can be manned (i.e., operated by an onboard human) or unmanned, and unmanned maritime vehicles can be remotely controlled or can be autonomous.

[0004] Maritime vehicles may be powered by a number of power sources such as, for example, fuel and/or fuel cells, electrical power sources, and the like. In arrangements where electrical power sources are used, it may be desirable to use rechargeable battery packs. However, regulatory requirements involving certification(s) to use such power sources may be difficult to satisfy and may take significant time to obtain. Additionally, such rechargeable power sources may require lengthy charge times for use, which in turn may result in significant fleet downtime.

[0005] Accordingly, there is a need for improved devices having improved functionalities.

SUMMARY

[0006] A retention assembly for retaining a power source in a maritime vehicle includes at least one strap member and a buckle. The at least one strap member includes a first end, a second end, and an elongated length therebetween. The buckle is operably coupled with a portion of the elongated length of the at least one strap member and includes a quick release mechanism movable between a first, disengaged position and a second, engaged position. The second, engaged position is adapted to generate a tension in the at least one strap member to retain the power source within a cavity.

[0007] In some examples, the retention assembly may further include a first plate that covers at least a portion of the power source. This first plate may be positioned adjacent to the at least one strap member and may absorb an external force exerted on the power source. In some approaches, the retention assembly may include a second plate that supports at least a portion of the power source. This second plate may include a wedged member.

[0008] In some forms, each of the first and second ends may include a mounting mechanism that operably couples the strap member with a structure member of the maritime vehicle. In some examples, the retention assembly may include a battery stand-off member that operably couples with the elongated length of the at least one strap member. The battery stand-off member may include a bracket member defining a slot to receive a portion of the at least one strap member. The battery stand-off member is adapted to exert a force on the power source when the quick release mechanism is in the second, engaged position. In these and other examples, the at least one strap member may include a positioning protrusion disposed on a portion of the elongated length thereof. This positioning protrusion may engage a groove formed in the battery stand-off member.

[0009] In some examples, the buckle may be in the form of a flip top quick-release buckle. In some examples, the at least one strap member may be constructed from a para-aramid material.

[0010] In accordance with a second aspect, a power system for a maritime vehicle including at least one support member includes a rechargeable power source disposed in a shell and a retention assembly for retaining the rechargeable power source. The retention assembly includes at least one strap member having a first end, a second end, and an elongated length between the first end and

the second end, and a buckle operably coupled with a portion of the elongated length of the at least one strap member. At least one of the first end or the second end of the at least one strap member is operably coupled with the at least one support member of the maritime vehicle. The buckle includes a quick release mechanism movable between a first, disengaged position, and a second, engaged position. The buckle is engageable to selectively retain the rechargeable power source within a cavity of the maritime vehicle and to release the rechargeable power source from the cavity of the maritime vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate examples of concepts that include the claimed invention, and explain various principles and advantages of those examples.

[0012] FIG. 1A is a top perspective view of an example of a maritime vehicle constructed in accordance with the teachings of the present disclosure;

[0013] FIG. 1B is a front view of the maritime vehicle of FIG. 1A;

[0014] FIG. 1C is a rear view of the maritime vehicle of FIG. 1A;

[0015] FIG. 1D is a bottom perspective view of the maritime vehicle of FIG. 1A;

[0016] FIG. 1E is another bottom perspective view of the maritime vehicle of FIG. 1A;

[0017] FIG. 1F is similar to FIG. 1A, but with the cap and various components of the maritime vehicle removed for illustrative purposes;

[0018] FIG. 1G is similar to FIG. 1F, but with additional components of the maritime vehicle removed for illustrative purposes;

[0019] FIG. 1H is a rear view of the maritime vehicle of FIG. 1F;

[0020] FIG. 1I is a first cross-sectional view taken along line I-I in FIG. 1A;

[0021] FIG. 1J is a second cross-sectional view taken along line J-J in FIG. 1A;

[0022] FIG. 1K is similar to FIG. 1A, but with the latching assembly of the maritime vehicle removed for illustrative purposes;

[0023] FIG. 2 is an exploded perspective view of an example power system for use with the maritime vehicle of FIGS. 1A-1K;

[0024] FIG. 3 is a cross-sectional perspective view of a portion of the example power system of FIG. 3; and

[0025] FIG. 4 is a close-up perspective view of a portion of the example power system of FIGS. 2 and 3.

[0026] Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions and/or relative positioning of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various examples. Also, common but well-understood elements that are useful or necessary in a commercially feasible examples are often not depicted in order to facilitate a less obstructed view of these various examples. It will further be appreciated that certain actions and/or steps may be described or depicted in a particular order of occurrence while those skilled in the art will understand that such specificity with respect to sequence is not actually required. It will also be understood that the terms and expressions used herein have the ordinary technical meaning as is accorded to such terms and expressions by persons skilled in the technical field as set forth above except where different specific meanings have otherwise been set forth herein.

[0027] The apparatus and method components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding examples of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

[0028] Although the figures show parts with clean lines and boundaries, some or all of these lines and/or boundaries may be idealized. In reality, the boundaries and/or lines may be unobservable, blended, and/or irregular. Use of terms such as up, down, top, bottom, side, end, front, back, etc. herein are used with reference to a currently considered or illustrated orientation. If they are considered with respect to another orientation, it should be understood that such terms must be correspondingly modified.

DETAILED DESCRIPTION

[0029] The present disclosure is directed to a maritime vehicle that is primarily intended for use for military purposes (e.g., for naval defense, patrolling waters and enforcing laws, reconnaissance, naval exploration, monitoring) but can also be used for other purposes if desired. The maritime vehicle is durable and configured to quickly, efficiently, and stealthily traverse a body of water once dispatched (e.g., from other maritime vehicles, beachheads, or an airdrop). The maritime vehicle is modular, with components that can be flexibly altered, removed, or added as desired in accordance with the mission of the maritime vehicle. The maritime vehicle can collaborate with other similar maritime vehicles and/or military assets when necessary. The maritime vehicle is preferably unmanned and autonomous though need not be.

[0030] FIGS. 1A-1K illustrate one example of a maritime vehicle **100** constructed in accordance with the teachings of the present disclosure. The maritime vehicle **100** is an unmanned vehicle configured to autonomously traverse a body of water. The maritime vehicle **100** generally includes a hull **104** and a cap **108** that is coupled to the hull **104** to secure various components within the maritime vehicle **100**. The hull **104** is at least partially disposed in the body of water in which the maritime vehicle **100** is traversing. The hull **104** is a mono-hull that has a front (or bow), a rear (or stern), two sides, and a keel coupled to another. The front, rear, sides, and the keel can be welded together or can be coupled to one another in a different manner. The hull **104** is configured such that the hull provides a continuous planing surface that allows the maritime vehicle **100** to be highly maneuverable and to ride along the top of a body of water at high speeds, even in extreme weather conditions and difficult to navigate bodies of water. Meanwhile, the cap **108** is coupled to the hull **104** to cover and/or conceal the components of the maritime vehicle **100** disposed in and carried by the hull **104** as the maritime vehicle **100** traverses the body of water.

[0031] In this example, the hull **104** and the cap **108** each have a length that is equal to approximately 6 feet. In other examples, however, the length can vary. For example, the length can be equal to approximately 14 feet. The hull **104** is preferably entirely made of aluminum but can be partially or entirely be made of fiberglass and/or one or more other materials. In other examples, the maritime vehicle **100** can include two or more hulls (e.g., two parallel hulls). In this example, the cap **108** entirely covers the hull **104** (and the components therein). In other examples, however, the maritime vehicle **100** need not include the cap **108** or the cap **108** may only partially cover the hull **104** (and the components disposed therein).

[0032] In some examples, the cap **108** can be removably coupled to the hull **104** via a locking system. For example, as illustrated in FIGS. 1A-1K, the cap **108** can be removably coupled to the hull **104** via a plurality of latch mechanisms disposed around a perimeter of the maritime vehicle **100**. Thus, the cap **108** can be removed to allow access to the interior of the hull **104**. In other examples, however, the cap **108** can be permanently coupled to the hull **104**.

[0033] The maritime vehicle **100** also includes a plurality of bulkheads **112** arranged within the hull **104**. The bulkheads **112** divide the maritime vehicle **100** into a plurality of different compartments for receiving and retaining different components in the maritime vehicle **100**.

[0034] The maritime vehicle **100** also includes a sensor system that is generally configured to collect data about various components of the maritime vehicle **100** as well as data about the environment surrounding the maritime vehicle **100** (including data about objects in that environment). To this end, the sensor system generally includes a plurality of sensors disposed on an exterior or an interior of the maritime vehicle **100**. The sensors can include, for example, one or more pressure sensors (e.g., positioned to detect the pressure of the ambient air external to the maritime vehicle **100**, the pressure of the water in which the maritime vehicle **100** is disposed, the pressure within the maritime vehicle **100**), one or more temperature sensors (e.g., positioned to measure a temperature of a component of the maritime vehicle **100**, a temperature of ambient air external to the maritime vehicle **100**, a temperature of water in which the maritime vehicle **100** is disposed), one or more acoustic sensors (e.g., sonar sensors), one or more LIDAR sensors, one or more location sensors (e.g., GPS sensors, compass sensors), one or more motion sensors (e.g., accelerometers, gyroscopes), one or more infrared sensors, one or more water level sensors, one or more humidity sensors, one or more power sensors (e.g., configured to detect charging or fueling levels), one or more lighting sensors (e.g., daylight sensors), one or more imaging sensors (e.g., CCD sensors, CMOS sensors), one or more magnetic sensors, or combinations thereof.

[0035] The maritime vehicle **100** also includes a power system **120** that is generally configured to power the maritime vehicle **100** (and the components of the maritime vehicle **100**). The power system **120** generally includes a propulsion system and one or more power sources **122** configured to power the propulsion system (and the other components within the maritime vehicle **100**). The propulsion system is generally configured to propel the maritime vehicle **100** in/on the water. The one or more power sources **122** can include, for example, one or more batteries, fuel (e.g., gasoline, diesel) stored in tanks carried by the maritime vehicle **100**, hydrogen stored in hydrogen tanks carried by the maritime vehicle **100**, solar panels (e.g., mounted to an exterior of the vehicle **100**), or other sources. The maritime vehicle **100** illustrated in FIGS. 1A-1K includes four battery assemblies each including a rechargeable battery. The maritime vehicle **100** generally also includes a cooling system configured to cool the propulsion system and/or the one or more energy sources, thereby preventing these components from overheating and leading to failure of the maritime vehicle **100**.

[0036] In operation, the maritime vehicle **100** may be used to deploy and/or retrieve payloads such as, for example, persons, weapons (e.g., drones, missiles, mines, bombs), cargo (e.g., food), scientific instruments, or other equipment. Payloads can be deployed aerially (into the air), underwater, or on the surface of the water. Payloads can also be retrieved from the air, from underwater, or the surface of the water. Payloads to be deployed can be disposed in the hull **104**, attached to the exterior surface of the hull **104**, or attached to the exterior surface of the cap **108** prior to deployment. Likewise, retrieved payloads can be stored in the hull **104**, attached to and stored on the exterior surface of the hull **104**, or attached to and stored on the exterior surface of the cap **108**.

[0037] The maritime vehicle **100** can also include other systems to help with the operation of the maritime vehicle **100**, namely a ballast system, a navigation system, and a vision system. The ballast system is generally configured to stabilize the maritime vehicle **100** in the water, regardless of whether the maritime vehicle **100** is stationary or on the move. To this end, the maritime vehicle **100** may include one or more ballast tanks or chambers selectively filled with water or air to vary the buoyancy of the maritime vehicle **100**. Alternatively, or additionally, the ballast system may include and utilize one or more inflatable devices to vary the buoyancy of the maritime vehicle **100**. The ballast system may also provide for the selective submerging and re-surfacing of the maritime vehicle **100** in a similar manner. The navigation system, which may for example be an inertial navigation system, utilizes the sensors of the sensor system to track the position and orientation of the maritime vehicle **100** and to guide the maritime vehicle **100** to its desired location in the body of water (or in a different body of water). The vision system is generally configured to

capture, process, and analyze images obtained by the one or more image sensors and other data (e.g., data obtained by other sensors in the sensor system). The vision system can in turn identify or classify the environment surrounding the maritime vehicle **100** (including objects in that environment). Finally, the payload control system is configured to deploy or retrieve payloads.

[0038] The maritime vehicle **100** further includes a communications system that is generally configured to facilitate communication (i) between the maritime vehicle **100** and one or more central (remote) controllers, (ii) between the maritime vehicle **100** and and/or one or more other maritime vehicles **100** and/or other military assets (e.g., planes, ships), and (iii) between different components of the maritime vehicle **100**. The communications system generally includes one or more local controllers and one or more communication modules (e.g., one or more antennae, one or more receivers, one or more transmitters, one or more radios, one or more ethernet switches) to effectuate wired or wireless communication between the maritime vehicle **100** and the central controller(s) or other maritime vehicles **100**. For example, the maritime vehicle **100** includes a plurality of antennae disposed on an exterior of the cap **108** as well as a plurality of antennae disposed in the hull **104**.

[0039] The one or more local controllers are generally configured to communicate data (e.g., operational instructions, data from the sensor system, data from other maritime vehicles **100** or military assets) and to perform automated operations of the maritime vehicle **100** based on that data. In some examples, the maritime vehicle **100** includes a plurality of different local controllers. For example, the maritime vehicle **100** can include one or more propulsion controllers (for controlling the operation of the propulsion system), one or more sensor controllers (for controlling the sensors in the sensor system), one or more payload controllers (for deploying or retrieving payloads), one or more navigation controllers (as part of the navigation system), and one or more ballast controllers (for controlling the ballast system). It will be appreciated that each of the one or more controllers may be implemented as hardware (e.g., processor, die, integrated device), software (e.g., non-transitory processor readable medium), and/or combinations thereof, in one or more devices (e.g., processor, chip, computer, tablet, mobile device).

[0040] While not explicitly described or illustrated herein, it will be appreciated that the maritime vehicle **100** includes several additional components. For example, the maritime vehicle **100** includes various sealing elements configured to provide seals between different components of the vehicle **100** (or between the vehicle **100** and the environment surrounding the vehicle **100**). As another example, the maritime vehicle **100** also includes various fasteners that help to couple the components of the maritime vehicle **100** together. As yet another example, the maritime vehicle **100** includes cabling that helps to communicatively couple components of the maritime vehicle **100** together. As yet another example, the maritime vehicle **100** includes various electrical components that help to operate the maritime vehicle **100**, e.g., one or more relay boards, one or more DC-DC converters.

[0041] With reference to FIGS. 2-4, aspects of the power system **120** will now be described. In these examples, the power system **120** incorporates a rechargeable power source **122** in the form of any number of batteries. More specifically, the rechargeable power source **122** is in the form of a lithium ion ("LI-ON") battery pack that may provide power to the propulsion system, the sensing systems, and/or any other desired systems or subsystems of the maritime vehicle **100**. Generally speaking, the power system **120** includes the rechargeable power source **122** and a retention assembly **130**. The rechargeable power source **122** may be disposed in a shell **124** that may provide structural rigidity and security from undesired objects (e.g., water, debris, etc.). It is to be appreciated that the rechargeable power source **122** may include any number of features to assist with providing power to components of the maritime vehicle **100**, and similarly may include any number of features to assist with charging the rechargeable power source **122** such as, for example, terminal posts, plugs, sockets, cables, and the like. Other examples are possible.

[0042] The rechargeable power source **122** is generally disposed within the hull **104** of the

maritime vehicle **100**. More specifically, the hull **104** may include any number of support members **106** disposed therewithin that may, along with the bulkhead(s) **112**, at least partially define cavities dimensioned to accommodate various components of the maritime vehicle **100**. In the illustrated examples, the maritime vehicle **100** includes four rechargeable power sources **122**; two of which are disposed on the port side of the maritime vehicle **100**, and two of which are disposed on the starboard side of the maritime vehicle **100**. However, other locations and/or arrangements may be used as desired.

[0043] The retention assembly **130** includes any number of strap members **132** having any number of mounting mechanisms **134** and a buckle **136**. More specifically, the strap member(s) **132** may have a first end **132a**, a second end **132b**, and an elongated length **132c** therebetween. In some forms, the strap member **132** may be constructed from a para-aramid fabric material having a high tensile strength, heat resistance, abrasion resistance, ballistic resistance, and/or chemical resistance.

[0044] In some examples, the mounting mechanism **134** is in the form of a bracket member adapted to couple with a desired support member **106** formed by or otherwise coupled with the hull **104**. In the illustrated examples, each of the first and second ends **132a**, **132b** of the strap member **132** includes a mounting mechanism **134**, but in other arrangements, only one of the first or the second ends **132a**, **132b** may have a mounting mechanism **134** coupled therewith. The mounting mechanism **134** may include an opening **134a** that receives a fastener **134b** (e.g., a bolt, screw, pin, or the like) that may operably engage a corresponding hole or opening formed in the support member **106**. It is to be appreciated that any number of desired approaches may be used to secure the mounting mechanism **134**, and thus the strap member **132**, with the support member **106**.

[0045] The buckle **136** includes a quick release mechanism **138** may be used to reduce and lengthen an overall length of the strap member **132**. More specifically, in the illustrated example, the quick release mechanism **138** is disposed along the elongated length **132c** of the strap member **132** such that a first intermediate end of the strap member **132** is coupled with a first section of the quick release mechanism **138** and a second intermediate end of the strap member **132** is coupled with a second section of the quick release mechanism **138**. The quick release mechanism **138** may be in the form of a flip-to-open buckle that is rotatable or otherwise pivotable between an engaged position and a disengaged position. More specifically, in the disengaged position, the quick release mechanism **138** effectively lengthens an overall length of the strap member **132** (due, at least in part, to portions of the quick release mechanism **138** being in an extended arrangement). Further, in the engaged position, the quick release mechanism **138** effectively reduces the length of the strap member **132** (due, at least in part to portions of the quick release mechanism **138** being in a folded, layered, or otherwise stacked arrangement). Put differently, in the disengaged position, the strap member **132** is loosened, and in the engaged position, the strap member **132** is tightened. In the illustrated examples, the quick release mechanism **138** is positioned in the engaged position.

[0046] The retention assembly **130** may include any number of additional components such as, for example, a first plate **140** and/or a second plate **144**. In the illustrated examples, the first plate **140** is positioned between a portion of the strap member **132** and the shell **124** of the rechargeable power source **122**, while the second plate **144** is positioned between the shell **124** of the rechargeable power source **122** and a floor portion of the hull **104**. Each of the plates **140**, **144** may be used to absorb shock and/or other impacts experienced by the power system **120** during operation. More specifically, in some examples, the first plate **140** may act to disperse the compressive or retention force exerted by the strap member(s) **132** on the rechargeable power source **122**, while the second plate **144** may be used to assist with securing and/or placement of the rechargeable power source **122** within the hull **104**. In these examples, the second plate **144** may be in the form of a wedge member that is secured with the hull (e.g., with one of the support members **106** and/or the bulkhead **112**) via bolts, rivets, and/or other suitable approaches. In the illustrated example, the wedge member assists with retaining the rechargeable power source **122** in a generally horizontal configuration.

[0047] The retention assembly **130** may also include a battery stand-off member **148** in the form of a bracket member that engages the shell **124** and the strap member **132**. More specifically, the stand-off member **148** is in the form of a body having a first planar surface **148a**, a second planar surface **148b**, and a slot **150**. The first planar surface **148a** and the second planar surface **148b** are generally perpendicular to each other such that they form a corner that receives a corresponding section of the shell **124**. It is to be appreciated that in some examples, the first and the second planar surfaces **148a**, **148b** may be positioned at any desired respective angle relative to each other. The slot **150** is dimensioned to receive a portion of the strap member **132** therethrough. In the illustrated examples, the strap member **132** includes a positioning protrusion **133** that is at least partially disposed within the slot **150** of the stand-off member **150** to provide alignment of these components relative to each other.

[0048] As a non-limiting example, portions of the power system **120** may be assembled as follows. First, the first end **132a** or the second end **132b** of the strap member **132** may be coupled with a support member **106** using a mounting mechanism **134**. The second plate **144** may be coupled with a support member **106** or bulkhead **112**, and the shell **124** of the rechargeable power source **122** may be placed on the second plate **144**. Next, the first plate **140** may be placed on the shell **124**, and the elongated length **132c** of the strap member **132** may be inserted into the slot **150** of the battery stand-off member **148**. In this arrangement, the buckle **136** is positioned above the first plate **140** and is initially in the disengaged position.

[0049] The battery stand-off member **148** may then be positioned on the corner of the shell **124**, and the first end **132a** of the strap member **132** may be coupled with a support member **106** via a mounting mechanism **134**. Last, the quick release mechanism **138** is closed or clasped such that the strap member **132** is shortened and tightened, thereby generating a compressive retention force on the rechargeable power source **122**. As previously noted, this compressive force is exerted on the first plate **140** in a dispersed manner, but additionally, the battery stand-off member **148** also provides a compressive retention force on the rechargeable power source **122** in the direction towards the hull **104**. These forces cooperate to retain the rechargeable power source **122** while absorbing any shock impacts experienced by the maritime vehicle **100**.

[0050] It is to be appreciated that in some examples, both ends **132a**, **132b** of the strap member **132** may first be coupled with support members **106** prior to assembly of the remaining components. In still other examples, the assembly process of the power system **120** may take other forms.

[0051] So arranged, the power system **120** described herein may provide for quick and efficient charging and replacement of the rechargeable power sources used by the maritime vehicle **100**. The power system **120** advantageously incorporates a quick-release retention assembly **130** that significantly reduces removal and installation times as compared with traditional battery mounting approaches. Notably, the retention system **130** described herein also ensures portions of the power system **120** may withstand shock loads of up to approximately 20 g (e.g., loads experienced by the maritime vehicle **100** from cresting up a wake or an impact from a combatant craft assault, for example). Such systems may be quickly implemented in large fleets of maritime vessels, and due to the removable nature of the battery sources, needn't undergo strenuous and lengthy certification processes that would otherwise delay implementation of such vessels into a desired fleet.

[0052] Finally, although certain maritime vehicles have been described herein in accordance with the teachings of the present disclosure, the scope of coverage of this patent is not limited thereto. On the contrary, while the invention has been shown and described in connection with various preferred embodiments, it is apparent that certain changes and modifications, in addition to those mentioned above, may be made. This patent covers all embodiments of the teachings of the disclosure that fairly fall within the scope of permissible equivalents. Accordingly, it is the intention to protect all variations and modifications that may occur to one of ordinary skill in the art.

[0053] In the foregoing specification, specific embodiments have been described. However, one of

ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present teachings. Numerous alternative examples could be implemented, using either current technology or technology developed after the filing date of this patent, which would still fall within the scope of the claims. Additionally, the described embodiments/examples/implementations should not be interpreted as mutually exclusive and should instead be understood as potentially combinable if such combinations are permissive in any way. In other words, any feature disclosed in any of the aforementioned embodiments/examples/implementations may be included in any of the other aforementioned embodiments/examples/implementations.

[0054] The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. The claimed invention is defined solely by the appended claims including any amendments made during the pendency of this application and all equivalents of those claims as issued.

[0055] Moreover, in this document, relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “has,” “having,” “includes,” “including,” “contains,” “containing” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a”, “has . . . a”, “includes . . . a”, “contains . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms “a” and “an” are defined as one or more unless explicitly stated otherwise herein. The terms “substantially”, “essentially”, “approximately”, “about” or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting example the term is defined to be within 10%, in another example within 5%, in another example within 1% and in another example within 0.5%. The term “coupled” as used herein is defined as connected, although not necessarily directly and not necessarily mechanically. A device or structure that is “configured” in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

[0056] Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, “A, B or C” refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, and (7) A with B and with C. As used herein, the phrase “at least one of A and B” is intended to refer to any combination or subset of A and B such as (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, the phrase “at least one of A or B” is intended to refer to any combination or subset of A and B such as (1) at least one A, (2) at least one B, and (3) at least one A and at least one B.

[0057] The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various examples for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed examples require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter may lie in less than all

features of a single disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter. [0058] Additionally, the benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. [0059] Finally, any references, including, but not limited to, publications, patent applications, and patents cited herein are hereby incorporated in their entirety by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

[0060] The patent claims at the end of this patent application are not intended to be construed under 35 U.S.C. § 112(f) unless traditional means-plus-function language is expressly recited, such as “means for” or “step for” language being explicitly recited in the claim(s). The systems and methods described herein are directed to an improvement to computer functionality, and improve the functioning of conventional computers.

[0061] Although certain example methods, apparatus and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all methods, apparatus and articles of manufacture fairly falling within the scope of the claims of this patent.

Claims

1. A retention assembly for retaining a power source in a maritime vehicle, the retention assembly comprising: at least one strap member having a first end, a second end, and an elongated length between the first end and the second end; and a buckle operably coupled with a portion of the elongated length of the at least one strap member, the buckle including a quick release mechanism movable between a first, disengaged position, and a second, engaged position; wherein the second, engaged position of the quick release mechanism is adapted to generate a tension in the at least one strap member to retain the power source within a cavity.
2. The retention assembly of claim 1, further including a first plate adapted to cover at least a portion of the power source, the first plate positioned adjacent to the at least one strap member and being adapted to absorb an external force exerted on the power source.
3. The retention assembly of claim 1, further including a second plate adapted to support at least a portion of the power source, the second plate comprising a wedged member.
4. The retention assembly of claim 1, wherein each of the first end and the second end thereof includes a mounting mechanism adapted to operably couple the strap member with a structural member of the maritime vehicle.
5. The retention assembly of claim 1, further including a battery stand-off member adapted to operably couple with the elongated length of the at least one strap member, the battery stand off member including a bracket member defining a slot to receive a portion of the at least one strap member, wherein the battery stand-off member is adapted to exert a force on the power source when the quick release mechanism is in the second, engaged position.
6. The retention assembly of claim 5, wherein the at least one strap member includes a positioning protrusion disposed on a portion of the elongated length thereof, the positioning protrusion adapted to engage a groove formed in the battery stand-off member.
7. The retention assembly of claim 1, wherein the buckle comprises a flip top quick-release buckle.
8. The retention assembly of claim 1, wherein the at least one strap member is constructed from a para-aramid material.
9. A power system for a maritime vehicle including at least one support member, the power system comprising: a rechargeable power source disposed in a shell; and a retention assembly for retaining the rechargeable power source, the retention assembly comprising: at least one strap member

having a first end, a second end, and an elongated length between the first end and the second end, wherein at least one of the first end or the second end is operably coupled with the at least one support member of the maritime vehicle; and a buckle operably coupled with a portion of the elongated length of the at least one strap member, the buckle including a quick release mechanism movable between a first, disengaged position, and a second, engaged position; wherein the buckle is engageable to selectively retain the rechargeable power source within a cavity of the maritime vehicle and to release the rechargeable power source from the cavity of the maritime vehicle.

10. The power system of claim 9, further including a first plate adapted to cover at least a portion of the rechargeable power source, the first plate positioned adjacent to the at least one strap member and being adapted to absorb an external force exerted on the power source.

11. The power system of claim 9, further including a second plate adapted to support at least a portion of the rechargeable power source, the second plate comprising a wedged member.

12. The power system of claim 9, wherein each of the first end and the second end thereof includes a mounting bracket adapted to operably couple the strap member with the at least one structural member of the maritime vehicle.

13. The power system of claim 9, further including a battery stand-off member adapted to operably couple with the elongated length of the at least one strap member, the battery stand off member including a bracket member defining a slot to receive a portion of the at least one strap member, wherein the battery stand-off member is adapted to exert a force on the power source when the quick release mechanism is in the second, engaged position.

14. The power system of claim 13, wherein the at least one strap member includes a positioning protrusion disposed on a portion of the elongated length thereof, the positioning protrusion adapted to engage a groove formed in the battery stand-off member.

15. The power system of claim 9, wherein the buckle comprises a flip top quick-release buckle.

16. The power system of claim 9, wherein the at least one strap member is constructed from a para-aramid material.
